

Cross domain tortuosity extraction methods

2026-09-27

Borrowing four fields to score coronary tortuosity

This report locks in a five-stage method for extracting a whole-tree tortuosity profile from a segmented, healthy branching vascular tree, built by deliberately splicing together validated pieces from four separate literatures rather than inventing a new one from scratch. Stage 1 keeps this project’s own chord-turning-angle curvature, κ_a , as the anchor but formalizes it inside the discrete Frenet-Serret apparatus and adds a matched torsion term, so that out-of-plane coiling is measured for the first time rather than silently folded into a curvature-only number. Stage 2 answers the project’s own unresolved smoothing-degree disagreement not by picking a side, but by sweeping the chord/smoothing scale and reporting a Richardson-Mandelbrot-style scale exponent alongside the fixed-scale value, so no single arbitrary resolution has to be defended. Stage 3 fuses two literatures that solve two different halves of the “whole tree” problem: retinal imaging’s length- and chord-weighted aggregation formulas (WAL and WAC), the only validated machinery anywhere in the four literatures for collapsing many unequal vessel segments into one number, and the Topological Morphology Descriptor (TMD) from computational neuroscience, which captures branch topology that the retinal formulas throw away entirely. Stage 4 lets the shape of each stage-3 output pick its own reference-range method: GAMLSS-style skew-aware centiles for the scalar aggregates, robust Mahalanobis distance on a Minimum Covariance Determinant (MCD) estimate for the fixed-length topological vector. Stage 5 argues for feeding a downstream risk model this hand-crafted vector rather than a learned tree embedding, on interpretability and small-sample grounds, while naming a learned embedding as the honest secondary comparison. None of this combination has been run or tested; the report closes by naming the single most important validation step, checking the transplanted aggregation formula’s compositionality inequalities and its correlation with expert grading on this project’s own coronary centerlines, exactly as the retinal source did before trusting its own formula, together with the other unresolved gaps the four literatures leave open. CGPS, the eventual out-of-distribution clinical cohort this project will one day apply the method to, is mentioned here only as that motivating horizon, never as a subject of this design.

Chord turning angles get a Frenet-Serret twin: torsion joins curvature

The project’s existing primary descriptor, κ_a , is built by resampling a centerline at a fixed chord length ℓ , forming unit chord directions $u_1, \dots, u_n$, and summing the turning angles $\theta_j = \arccos(u_j \cdot u_{j+1})$ between consecutive directions, normalized by arc length L . This is not an ad hoc formula: it is exactly the discretization that the differential-geometry literature uses for total curvature of a curve, Milnor’s own definition of total curvature for a possibly non-smooth curve is the supremum, over all inscribed polygons, of the sum of the polygon’s exterior turning angles ([Milnor, total curvature of knots](#); [six proofs of the Fáry-Milnor theorem](#)), and the continuous Frenet-Serret frame

that κ_a is discretizing evolves by $dT/ds = \kappa(s)N(s)$, $dN/ds = -\kappa(s)T(s) + \tau(s)B(s)$, $dB/ds = -\tau(s)N(s)$, with $\kappa(s)$ and $\tau(s)$ the curvature and torsion at arc-length position s ([space curves and Frenet frames](#)). For closed curves, this same total-curvature quantity has hard topological floors, Fenchel’s theorem bounds it below by 2π for any simple closed space curve, and the Fáry-Milnor theorem raises that floor to 4π for any nontrivially knotted closed curve ([Fenchel and Fáry-Milnor summary](#)); an open coronary branch has no such closed-curve theorem to invoke, but the point stands that κ_a is measuring a quantity the mathematics literature treats as fundamental, not an arbitrary index invented for vessels.

What κ_a cannot see is torsion, the rate at which the curve’s osculating plane itself rotates, $\tau(s)$ in the equations above. Two branches can have identical total curvature while one bends entirely within a single plane and the other spirals in three dimensions, a helical, corkscrew pattern that visual tortuosity grading in the cerebrovascular literature already treats as a distinct category worth naming separately from ordinary bending ([novel vessel tortuosity scale, large vs. small turns](#)). This design adds a discrete torsion companion to κ_a using the same chord-resampled polyline: the discrete Frenet frame construction defines the unit tangent $T_i = (c_{i+1} - c_i)/|c_{i+1} - c_i|$, the discrete normal $N_i = (T_i - T_{i-1})/|T_i - T_{i-1}|$, and the discrete binormal $B_i = T_i \times N_i$, so that torsion is read off as the turning of the (N, B) plane between consecutive frames ([Frenet-Serret estimation of curvature and torsion](#)). Writing ϕ_j for that binormal-plane turning angle at chord ℓ , the mean absolute torsion is defined by direct analogy with κ_a , $\tau_a = \sum_j |\phi_j|/L$, using the same chord length and the same arc-length normalization already locked in for κ_a , so that the two numbers are directly comparable and reuse the identical resampling machinery rather than introducing a second, incompatible discretization. As a robustness cross-check on the turning-angle curvature itself, the three-point Menger curvature, the reciprocal of the circumradius through three consecutive points, $c(x_1, x_2, x_3) = 4\Delta/(abc) = 1/R$ for triangle area Δ and side lengths a, b, c ([Menger curvature](#)), gives a coordinate-free, closed-form alternative discretization of the same local curvature that can be computed on the identical point set without needing chord resampling at all; it is proposed here only as a sanity check on κ_a ’s discretization choice, not as a replacement, because the turning-angle form is what the project’s own reliability work has already characterized.

Richardson plots replace the single smoothing choice

The single most consequential open disagreement this project already carries is how much extra smoothing to apply before computing κ_a : one internal report locks $\sigma = 0$ beyond the skeleton’s own 0.5 mm/5-vertex pass, a later report recommends an extra $\sigma = 1$ mm, and the readiness audit names this disagreement explicitly as unresolved and blocking. The cross-domain literature offers a way out of that argument that neither side of it considered: stop treating smoothing degree as a single hyperparameter to be won or lost, and instead measure how the descriptor behaves as a function of scale, then report a scale-robust summary of that function. The master template for this is Richardson’s empirical observation, formalized by Mandelbrot, that a jagged curve’s measured length grows as the measurement step ℓ shrinks according to $L(\ell) \propto \ell^{1-D}$, where D is a non-integer roughness exponent between 1 and 2 for a plane curve, so that a log-log plot of L against ℓ has slope $D - 1$ ([Mandelbrot, how long is the coast of Britain](#)). This same relation is the acknowledged ancestor of at least three other scale-dependence tools found across the four literatures: a scale-specific sinuosity metric for river meanders explicitly derived from the Richardson plot, reported to exist in the geomorphology literature though its exact governing equation could not be verified beyond the abstract level in this project’s research pass ([scale-specific sinuosity S3](#)); Higuchi’s fractal-dimension estimator, which constructs lagged sub-series of a 1D signal and fits an analogous

power law between average curve length and lag k ([Higuchi dimension](#)); and curvature scale space (CSS), which convolves a curve’s coordinate functions with Gaussians of increasing width σ and recomputes curvature at every scale, so that the number of curvature zero-crossings (real inflection points) shrinks monotonically as σ grows, and a curve’s shape is represented by the whole family of zero-crossing locations across scale rather than one smoothing choice ([Mokhtarian and Mackworth, curvature scale space](#)), later adopted as an MPEG-7 shape standard ([CSS and MPEG-7](#)) and extended to 3D via torsion scale space and a combined generalized torsion-curvature scale space descriptor, the latter’s exact equations not independently verified in this pass ([torsion scale space](#); [GTCSS, abstract level](#)).

This design adopts the Richardson-plot route rather than full Gaussian CSS for the primary scale-robustness mechanism, for a concrete reason tied to what the project already has: κ_a is already computed at a fixed chord ℓ , so sweeping ℓ over a range, say from a lower bound set by the centerline’s own point spacing up to an upper bound set by the shortest clinically meaningful segment, reuses the existing chord-resampling code path exactly, with no new smoothing kernel to implement or justify. Concretely, computing $\kappa_a(\ell)$ (equivalently, computing $L(\ell)$, the apparent arc length at chord ℓ) at several values of ℓ and fitting $\log \kappa_a(\ell)$ against $\log \ell$ gives a slope, the Richardson exponent D_κ , that is by construction a property of the whole scale-dependence curve rather than of any single ℓ . D_κ is added as a companion feature alongside the fixed-scale κ_a measured at the project’s own already-vetted $\ell = 5$ mm operating point, rather than replacing it: the project’s own case study found κ_a ’s cohort-percentile ranking barely moved between $\sigma = 0$ and $\sigma = 1$ mm smoothing, which is direct evidence that the fixed-scale anchor is already reasonably stable and does not need to be discarded, while D_κ adds exactly the scale-behavior information a fixed-scale scalar cannot carry. Full Gaussian CSS is named here as the natural next step if a diagnostic, feature-level view of exactly which bends survive across scale is later needed (an analogue to the inflection-count metric already used in the cerebral literature), but it requires a genuinely new smoothing/differentiation pipeline the project does not yet have, so it is not locked into this design’s required path.

A distinct, complementary answer to “which scale” comes from the CAD and computer-graphics curvature-estimation literature, which reframes the problem as global regularization rather than filtering: methods that minimize a discrete total-curvature or total-variation-of-the-normal functional recover sharp features while suppressing noise, with the sharp/smooth tradeoff controlled by a single regularization weight rather than a smoothing bandwidth applied before differentiation ([minimizing discrete total curvature, CVPR 2020](#)); a separate, more radical position argues that when scale itself is the unwanted nuisance variable, the principled fix is not to select a scale at all but to engineer the descriptor so a scale change becomes a symmetry the representation is invariant to, for instance via log-polar sampling followed by a Fourier-modulus transform ([scale invariance without scale selection, Kokkinos and Yuille](#)). This design does not adopt either as the primary mechanism, both trade the smoothing-degree argument for a different, equally unresolved argument, choosing a regularization weight or a transform domain, and neither has a verified equation or selection rule in this project’s research pass; they are named as the honest alternative route a future iteration could take if the Richardson-exponent sweep turns out to be insufficiently discriminating in practice.

Weighted retinal aggregation turns segments into one tree score

Stage 2 produces, per vessel segment, a small vector of scale-anchored and scale-swept scalars: κ_a , τ_a , TI , and D_κ . Turning many such per-segment vectors, one per named coronary branch, into a

single whole-tree score is exactly the problem retinal-imaging tortuosity research has already solved with the most explicit, tested machinery found anywhere across all four literatures surveyed for this report. Given N vessel sections with local tortuosity $\tau(C_i)$, length $L(C_i)$, and chord $D(C_i)$, one retinal-imaging source defines and empirically tests five combination schemes: an unweighted mean, Weighted Additivity on Length ($\tau_{WAL} = \sum_i L(C_i)\tau(C_i) / \sum_i L(C_i)$), a length-weighted mean-of-means variant, Weighted Additivity on Chord ($\tau_{WAC} = \sum_i D(C_i)\tau(C_i) / \sum_i D(C_i)$), and a global density variant ([bridging local and global tortuosity of retinal vessels](#)). Critically, this is not just five formulas offered neutrally: the same source validates them two ways, first mathematically, against the Hart inequality, $\min(\tau(C_1), \tau(C_2)) \leq \tau(C) \leq \max(\tau(C_1), \tau(C_2))$ for a vessel split into two sub-segments, and the stricter Grisan inequality for three-way splits, and second empirically, against ophthalmologist rankings on the BioImLab RET-TORT dataset, where the length-weighted (WAL) and chord-weighted (WAC) schemes combined with total-curvature-over-length or total-curvature-over-chord local indices achieved the highest correlation with expert judgment, Spearman $\rho \geq 0.860$, clearly outperforming a naive unweighted mean ([same source](#)). This is a deliberate, chosen precedent, not one option among many read neutrally: no other literature surveyed here, not cerebral, not carotid, not renal, not river geomorphology, not fractal-dimension work, produced a validated formula for fusing multiple unequal vessels into one tree-level number; the cerebral and carotid literature instead reports panels of separate named-vessel values compared by group statistics, and the Marfan visceral-artery literature explicitly declines to fuse its measured arteries into one score at all, reporting each separately instead ([Marfan visceral artery tortuosity panel](#)). The retinal WAL/WAC formulas are therefore locked in as this design’s aggregation mechanism specifically because they are the only aggregation machinery in the entire cross-domain search that has been checked against both a mathematical self-consistency criterion and a human-perception ground truth.

The choice of which weighting to pair with which local scalar is not arbitrary either. κ_a is already a per-unit-length quantity, total turning divided by arc length, so weighting the tree-level average by length again reflects how much vessel material carries that value, not a redundant second length correction; this design therefore defines $\kappa_{a,\text{tree}} = \sum_i L_i \kappa_{a,i} / \sum_i L_i$, a WAL-style aggregate, directly mirroring the retinal literature’s own best-performing pairing of a length-normalized local index with length weighting. TI , by contrast, is intrinsically defined through the chord distance D ($TI = L/D - 1$), so its natural aggregation partner is the chord-weighted WAC form, $TI_{\text{tree}} = \sum_i D_i TI_i / \sum_i D_i$, mirroring the retinal source’s own finding that a chord-normalized local index paired with chord weighting was likewise among the top performers. The Hart inequality is satisfied by construction for any convex (positively weighted, weights summing to the total) combination of the constituent values, but this design treats that as a property to be checked on this project’s own centerlines rather than assumed, precisely because “satisfied by construction” for one splitting convention does not guarantee it survives the project’s own, separately noted ambiguity between two different definitions of what counts as one named vessel (`tortuosity/vessels.py`’s longest-geodesic-path convention versus `topology.graph.Vessel`’s segment-chaining convention); this design inherits the project’s already-locked choice of the former for exactly this reason, since it is the definition every cited internal normal-range number was actually computed under, and because a WAL/WAC aggregate is only as trustworthy as the segmentation-into-vessels it is aggregating over. The left main, per the project’s own already-locked practice, is excluded from this weighted sum and reported separately by length and bifurcation angle wherever κ_a is undefined at the chosen chord, rather than silently dropped or forced into the aggregate with a NaN treated as zero.

Persistence diagrams add the branching topology WAL/WAC discards

The WAL/WAC aggregate has a structural blind spot worth naming plainly: it treats a tree as a bag of independent vessel scalars weighted by size, and discards everything about how those vessels connect, how many generations of branching occur, how asymmetric the branch structure is, which is exactly the kind of information the cerebral-tortuosity literature’s own finding warns cannot be assumed irrelevant, a single tortuosity index’s correlation with aneurysm risk held for the MCA in one study but not for the other named vessels tested, suggesting branch-role and branch topology genuinely matter and are not fully captured by a length-weighted scalar alone ([novel vessel tortuosity scale](#)). Computational topology offers a way to fold topology and geometry together into one object rather than treating them as two disconnected feature families, and the specific tool this design locks in is the Topological Morphology Descriptor (TMD), originally built for scoring whole neuronal dendritic trees. TMD assigns a real-valued descriptor function f to every point of the tree, sweeps the sublevel sets $T_t = \{x \in \text{tree} : f(x) \leq t\}$ as t increases, and records a persistence point $(f(m_2), f(s))$ each time two growing components merge at a branch point s , where m_1, m_2 are the two merging components’ extrema and $f(m_1) < f(m_2)$; the resulting collection of persistence points, one per branch, is a single persistence diagram D_{gf} describing the whole tree at once ([Kanari et al., metrics for comparing neuronal tree shapes based on persistent homology](#), primary source verified in full text). The same source proves this diagram is strictly more informative than the classical single-function tree-shape statistic (Sholl analysis, the count of tree crossings of a sphere of radius r centered at the root), showing the Sholl profile can be exactly recovered by counting persistence points in specific quadrants, which is direct, primary-source-verified evidence that a persistence diagram is not merely a different way of summarizing a tree but a strictly richer one.

This design chooses the geodesic path length from the ostium, arc length measured along the vessel rather than Euclidean straight-line distance from the root, as the descriptor function f , because coronary branches curve substantially and a straight-line distance from the ostium would not track how many branch generations deep a point actually sits, while path length does; this also reuses the identical arc-length parametrization already locked in for κ_a and τ_a at stage 1, so the whole pipeline shares one consistent notion of distance along the vessel rather than introducing a second, incompatible one. To obtain a fixed-length feature usable downstream, the diagram is converted to a smooth density $\rho_D(x) = \sum_i m_i K_t(x_i, x)$, weighting each persistence point $p_i = (x_i, y_i)$ by its lifetime $m_i = |y_i - x_i|$ under a Gaussian kernel K_t of bandwidth t , then discretized on an evenly spaced grid into an m -dimensional vector ([same source, verified in full text](#)). TMD is chosen over the two other whole-tree topological alternatives found in this research pass for a stated reason in each case. Against merge-tree edit distances, which are proven metrics with an explicit, primary-source-verified cost model and a stability bound relating them to the harder-to-compute interleaving distance ([Pegoraro, finitely stable edit distance for merge trees](#)), TMD is preferred here because it already comes with an explicit vectorization route into a fixed-length feature suitable for direct use in a downstream ML model, whereas the merge-tree edit-distance literature retrieved in this pass stops at pairwise distances between trees, with population-level outlier scoring documented only at the search-summary level and not independently verified. Against BHV (Billera-Holmes-Vogtmann) tree space, a genuinely different mechanism that represents edge-weighted trees as points in a CAT(0) orthant complex admitting well-defined Fréchet means and PCA directly on trees ([BHV tree space statistics](#)), TMD is preferred because BHV presumes a common, labeled leaf/branch correspondence across the trees being compared, a natural assumption in phylogenetics with a fixed species set, but not a natural one here, since coronary trees vary substantially between patients in exactly how many visible side branches are captured, and BHV would require solving

that branch-correspondence problem before any comparison could begin, an extra step TMD’s construction avoids entirely.

GAMLSS for the scalar, robust Mahalanobis for the vector

The reference-range method for stage 4 is chosen after, and because of, what shape of object stage 3 hands it, not before. $\kappa_{a,\text{tree}}$ and TI_{tree} are scalars, one number per patient, and the retinal-imaging literature’s own reporting convention is instructive here: tortuosity-index distributions from computer-assisted programs are described as typically positively skewed, which is the stated reason that field prefers medians and interquartile ranges over means and standard deviations, a claim that is itself only search-summary-level in the underlying research notes and flagged as such rather than independently verified against a primary source, and the one large population-based normative study located in this research pass, built from 10,151 UK Biobank participants, likewise reports its six whole-tree tortuosity sub-metrics as means with standard deviations alongside medians with interquartile ranges, stratified by sex and age decade, rather than assuming Gaussian symmetry outright ([population-based normative reference for retinal microvascular atlas](#); that source’s own exact formulas for its six sub-metrics were not disclosed beyond a supplementary table this pass could not retrieve, so only its reporting convention, not its equations, is drawn on here). For a skewed scalar like $\kappa_{a,\text{tree}}$ or TI_{tree} , this design locks in GAMLSS rather than the simpler LMS method it generalizes, specifically because LMS commits in advance to a single Box-Cox-normal (BCCG) family, whereas GAMLSS allows the full distribution, location, scale, and shape parameters, to each be modeled as smooth functions of a covariate from a broad family that includes the Box-Cox Power Exponential (BCPE) distribution ([GAMLSS and BCPE](#); [BCPE parameterization](#)); since this project has not yet measured the exact skew and kurtosis of a tree-level $\kappa_{a,\text{tree}}$ distribution, committing to LMS’s fixed family ahead of that measurement would repeat exactly the kind of premature single-choice commitment this whole design is built to avoid at the scale-selection stage. Quantile regression is named as the fallback if even GAMLSS’s broad family proves too restrictive, since it makes no distributional assumption at all, fitting each target quantile directly, at the cost of needing roughly 119 cases for a stable 90 percent confidence interval around a 95 percent reference interval and treating each quantile somewhat independently rather than sharing information across them ([quantile regression reference intervals](#)).

The TMD-derived density vector is a different shape of object entirely, a fixed-length multivariate vector rather than a scalar, and none of LMS, GAMLSS, or quantile regression is inherently multivariate; this is precisely the point at which the design must switch reference-range mechanisms rather than force one method to cover both outputs. The standard multivariate answer is Mahalanobis distance from a reference population’s mean and covariance, but the plain (classical) version is not safe to use here: it is well documented that a small number of outlying or mis-segmented reference cases can arbitrarily distort the sample mean and covariance it depends on, since the classical estimator’s breakdown point is effectively zero ([Mahalanobis distance sensitivity to outliers](#)). This design locks in the robust alternative instead: the Minimum Covariance Determinant (MCD) estimator finds the h -observation subset of the reference population whose covariance matrix has the smallest determinant, and uses that subset’s mean and covariance in place of the full-sample plug-in estimates, raising the effective breakdown point to roughly $(n - h)/n \approx 50\%$ for the default choice of h ([MCD breakdown point](#)), computed efficiently via the FAST-MCD algorithm’s iterative C-steps ([Rousseeuw and Van Driessen FAST-MCD](#)). This choice is not a theoretical nicety here: the reference population this design would actually draw on at this project’s current stage is its own curated ImageCAS-X cohort, and any undetected segmentation or centerline-extraction error in even

a small minority of that reference set would silently distort a classical Mahalanobis reference range in exactly the way MCD is built to resist. A directly relevant precedent for this whole pattern, computing a normative Mahalanobis-type distance along an anatomical curve’s own multivariate profile, already exists outside vascular tortuosity work: Mahalanobis Distance Tractometry compares an individual’s diffusion-tensor profile along a white-matter tract against a normative healthy-control sample, structurally the same problem as scoring one patient’s tree-level topological vector against a reference population’s ([MaD-Tract](#)).

Hand-crafted features first, learned embeddings as a check

The output this pipeline hands to a downstream risk-prediction model is a fixed-length, hand-crafted vector per patient tree: $\kappa_{a,\text{tree}}$, TI_{tree} , $\tau_{a,\text{tree}}$ computed the same way, the Richardson exponent D_κ , and the TMD density vector, each with its own reference-range or anomaly-distance transform already applied from stage 4. This is a deliberate choice against the alternative most directly available in the graph-learning literature, an end-to-end learned tree or graph embedding, computed by a graph neural network with a permutation-invariant readout (sum, mean, or an attention-based pooling such as Set2Set, mechanisms that ingest a graph of any size or branching structure by construction, [GNN readout mechanisms](#)) or a recursive tree-autoencoder with weight-tied branch operations that walks the tree structure explicitly, a mechanism already demonstrated for generative modeling of branching vessel geometry, though its exact architectural detail could not be verified beyond the abstract level in this research pass because the full text was access-blocked ([recursive VAE for vessel generative modeling, abstract level](#)). The radiomics methodology literature gives a direct, primary-source-titled reason to prefer hand-crafted features at the sample sizes typical of a thesis-scale clinical cohort: one comparison paper’s own title states plainly that deep features from pretrained networks do not outperform hand-crafted features in radiomics ([hand-crafted vs. deep radiomics](#)), a companion source states that deep feature extraction requires a large database to avoid overfitting while hand-crafted extraction does not carry that constraint ([handcrafted vs. deep learning radiomics, Lancet Digital Health](#)), and a further caution states explicitly that deep learning’s known reproducibility and overfitting problems are amplified, not merely unhelped, at small sample sizes ([deep learning amplifies small-N overfitting](#)). Hand-crafted features also carry a second advantage specific to this project: every one of the five components in the final vector is individually traceable back to a named clinical or geometric quantity a reader or clinician can inspect and sanity-check, which is exactly the property the explainable-AI literature credits hand-crafted features with and finds latent deep features to lack ([handcrafted vs. deep, interpretability](#)). Existing graph-neural-network work on coronary trees specifically, moreover, targets anatomical branch labeling and segmentation, not a risk-prediction embedding, so there is no existing precedent to repurpose even if a learned-embedding arm were attempted ([GNN coronary tree labeling](#)); this design names a learned tree embedding as the honest secondary, exploratory comparison arm once sample size supports it, consistent with the moderately supported (search-summary level, not independently verified against primary text) finding that a hybrid of hand-crafted and learned features tends to outperform either alone, but does not adopt it as the primary route now.

What would have to be validated before this method can be trusted

Every individual mechanism assembled above has been validated somewhere, by someone, on some population: WAL/WAC against ophthalmologist grading on retinal images, TMD against the Sholl-analysis special case on neuronal trees, MCD’s breakdown-point guarantee in the general robust-statistics literature, GAMLSS on WHO growth references. None of that validation transfers auto-

matically to the specific combination proposed here, and this report states that plainly rather than implying otherwise: this is a design proposal that splices validated pieces from four unrelated fields into a pipeline that, as a whole, has never been run on a single coronary tree. The most important validation step before any number produced by this pipeline could be trusted is to repeat, on this project’s own coronary centerlines, exactly the two-part check the retinal source performed before trusting its own aggregation formula: verify the Hart and Grisan compositionality inequalities directly on real coronary vessels split into sub-segments (confirming $\min_i \kappa_{a,i} \leq \kappa_{a,\text{tree}} \leq \max_i \kappa_{a,i}$ actually holds under this project’s own vessel-splitting convention and is not merely guaranteed by the arithmetic of a convex combination in the abstract), and separately measure the correlation between the resulting $\kappa_{a,\text{tree}}$ and expert visual tortuosity grading on a sample of this project’s own scans, the way the RET-TORT correlation of $\rho \geq 0.860$ was established for retinal images before that formula was trusted for retinal use ([bridging local and global tortuosity](#)). Without that check, there is no evidence that a formula validated on retinal vascular trees behaves sensibly at all on a structurally different vascular bed with far fewer, much larger, and far more clinically consequential branches.

Several further gaps carry forward as named, specific risks rather than vague hedging. No field examined outside retinal imaging, not cerebral, not carotid, not renal, has published a validated whole-network sinuosity or tortuosity aggregate; the WAL/WAC transplant onto a coronary tree is therefore a genuinely novel combination with no direct precedent even within the vascular-imaging literature broadly, only within its retinal sub-field. Elastic shape analysis via the square-root velocity function (SRVF) is, on its own terms, the most mathematically rigorous single-curve shape metric found in this research pass, fully invariant to translation, rotation, and reparametrization, and supporting a genuine Fréchet mean and population variability structure ([SRVF elastic shape analysis](#)), but no confirmed construction of an SRVF-type mean shape or normal range for a whole tree of curves, as opposed to a single curve or a landmark point set, was found anywhere in the retrieved literature; this is exactly why SRVF is not used as this design’s backbone despite its mathematical elegance, and it is named here as an open problem for a future iteration to solve, not quietly worked around. The direction in which fractal-dimension-style complexity measures move under disease or disturbance is not universal across the fields that use them: urbanization raises drainage-network fractal dimension, soil compaction raises root-system fractal dimension, but drought stress raises it in some plant species, lowers it in others, and leaves it unchanged in others still, with no single governing rule ([fractal dimension direction under disturbance, root systems](#)). This design deliberately refuses to assume, ahead of measurement, that a diseased coronary tree’s Richardson exponent D_κ will come out higher or lower than a healthy one’s; that has to be measured on real data, not asserted by analogy. Finally, several specific formulas referenced above for their qualitative logic rather than their exact mathematics, the UK Biobank study’s six whole-tree tortuosity sub-metrics, the S3 scale-specific sinuosity metric, and the generalized torsion-curvature scale space descriptor, remain at the abstract or search-summary level in the underlying research and were never independently verified against primary text; none of their exact equations are used verbatim anywhere in this design, precisely because they cannot yet be trusted to that level of precision, and any future attempt to adopt their exact forms rather than their general logic would need a dedicated full-text read first.

The overall shift this design proposes is away from treating tortuosity as a single scalar computed at a single scale on a single vessel, the operating point the project has, reasonably, defended for its own good reasons so far, and toward treating it as a small, structured feature vector whose scale-dependence, whole-tree aggregation, and topological context are each handled by a purpose-built method borrowed from whichever field already solved that specific sub-problem best. The result

is not a single “more sophisticated” tortuosity number to replace κ_a ; it is a five-stage architecture in which κ_a keeps its existing, already-defended role as the scale-anchored primary scalar, and everything else, torsion, the Richardson exponent, the weighted tree aggregate, the topological vector, and the two matched reference-range methods, is new structure built around it rather than in place of it.